

App Dev Project Report

1. Student Details

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About Me: I am currently pursuing BS Degree program at IIT Madras and am also enrolled in the B.Tech in Computer Science and Data Science Engineering from VIT Pune . My interests lie in software engineering, backend development, and data-driven systems. I enjoy building full-stack applications and designing scalable systems that solve real-world problems. This project reflects my focus on backend architecture, database design, and practical web application development using technologies such as Flask and SQLite.

2. Project Details

Project Title: Placement Portal Application

Problem Statement: The objective of this project is to design and implement a web-based placement portal that automates the placement workflow and provides role-based access for students, companies, and administrators.

Approach: The system was developed using Flask (Python) as the backend and SQLite as the database. The frontend uses HTML, Bootstrap, and Jinja templates for dynamic rendering. The application implements role-based access for Admin, Company, and Student users, with authentication handled using Flask-Login and features such as job posting, application tracking, and dashboards.

3. AI/LLM Declaration

I used **ChatGPT (GPT-5)** for **guidance, troubleshooting, and improving code structure**.

The extent of AI/LLM usage is around **8-10 %**.

All final implementation **logic, debugging, and integration were done manually**.

4. Technologies and Frameworks Used :

Technology / Library	Purpose
Flask	Core backend web framework
SQLite	Lightweight database used to store application data
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap 5	Frontend styling and responsive UI
Chart.js	Data visualization (charts, statistics dashboards)
Flask-Login	User authentication and session management

5. Database Schema / ER Diagram

Database Tables :

Table	Description
Student	Stores student profile details including name, email, password, phone number, skills, and uploaded resume.
Company	Stores company registration details such as company name, email, industry, location, approval status, and authentication credentials.
Admin	Stores administrator login credentials used to manage the placement portal.
Placement Drive	Stores job or placement drive details posted by companies including job title, description, required skills, salary range, and application deadline.
Application	Stores student applications submitted for placement drives, including application status and submission time.
Placement	Stores final placement records when a student is successfully selected for a job.

Database Relationships :

➤ Company → PlacementDrive → One-to-Many

One company can post multiple placement drives.

➤ Student → Application → One-to-Many

One student can submit multiple job applications.

➤ PlacementDrive → Application → One-to-Many

One placement drive can receive applications from multiple students.

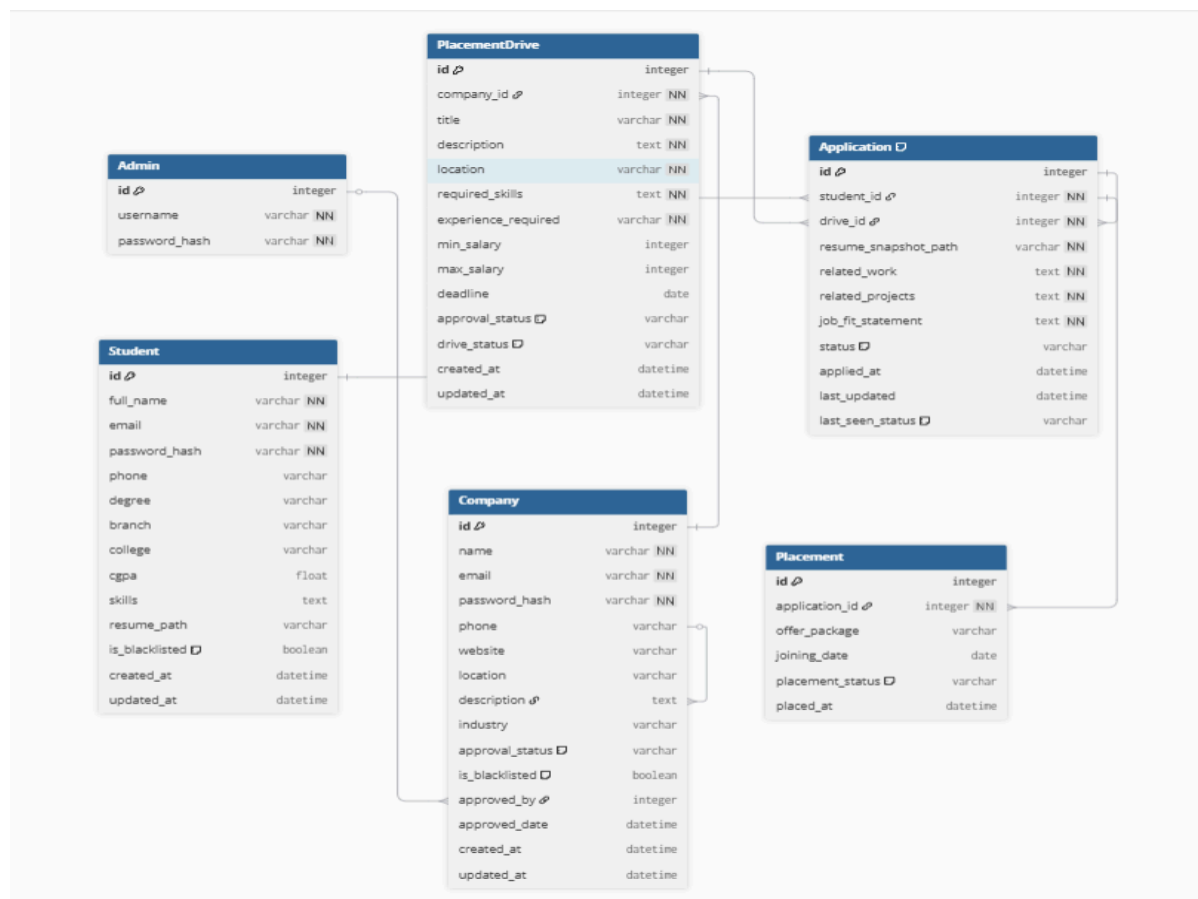
➤ Application → Placement → One-to-One

Each selected application results in one placement record.

➤ Company → Application → One-to-Many (Indirect through PlacementDrive)

A company can receive multiple applications through its placement drives.

ER Diagram :



6. API Resource Endpoints.

1. Common Routes

Endpoint	Method	Description
/	GET	Landing / Home page
/login	GET, POST	Login user (Student / Company / Admin)
/logout	GET	Logout and clear session
/student/register	GET, POST	Register a new student
/company/register	GET, POST	Register a new company

2. Student Module

Endpoint	Method	Description
/student/dashboard	GET	View student dashboard with available drives
/student/profile	GET, POST	View or update student profile
/upload_resume	POST	Upload student resume
/apply_drive/<drive_id>	POST	Apply to a placement drive
/student/applications	GET	View all applications submitted by the student

3. Company Module

Endpoint	Method	Description
/company/dashboard	GET	View company dashboard
/post_drive	GET, POST	Create a new placement drive

<code>/view_job_applications/ <drive_id></code>	GET	View applications for a specific drive
<code>/toggle_drive_status/ <drive_id></code>	POST	Activate or close a placement drive
<code>/update_application_status/ <application_id>/<status></code>	POST	Update application status (Shortlisted / Interview / Rejected / Placed)

4. Admin Module

Endpoint	Method	Description
<code>/admin/dashboard</code>	GET	View overall system statistics
<code>/admin/manage_students</code>	GET	View all registered students
<code>/admin/manage_companies</code>	GET	View all registered companies
<code>/admin/manage_jobs</code>	GET	View all placement drives
<code>/admin/manage_applications</code>	GET	View all job applications
<code>/blacklist_student/<student_id></code>	GET	Blacklist a student

7. Architecture and Features

Architecture Overview :

app.py – Main Flask application containing routes, authentication logic, and database operations.

/templates – Jinja2 HTML templates used for rendering pages such as dashboards, login, and management pages.

/static – Contains static assets such as images used for login pages and dashboards.

SQLite Database – Stores data related to students, companies, placement drives, and applications.

Session Management – Implemented using Flask session and Flask-Login for authentication.

Additional Features

- **Resume Version Locking:** If a student is **shortlisted or selected for an interview**, the resume used during that application remains fixed for that company. Any later resume updates will only be visible to **new companies or future drives**.
- **Editable Student Profile:** Students can update their **profile details and resume at any time**, except for applications already in advanced stages.
- **Placement Restriction:** Once a student is **marked as Placed**, they are automatically restricted from applying to further placement drives.
- **SPC Form Before Application:** Before submitting a final application, students must fill an **SPC (Student Profile for Company) Google Form**. Companies review this information first to check **relevant skills and experience** related to the job description before deciding whether to view the full resume.
- **Administrative Controls:** Admin can **blacklist students** and monitor platform activity through the admin dashboard.

8. Video Presentation

Drive Link:

https://drive.google.com/file/d/1OR0Q8yGo5sEdkyNoa_sVNYReA1PdxfJ/view?usp=sharing